Project Report: Milestone 4

NETFLIX

SOFTWARE ENGINEERING

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Objective:

The objective of this milestone is to analyse and model the Netflix streaming system using structured and object-oriented design techniques. This phase focuses on representing how data flows, processes, and interactions occur within the system. By developing logical models such as the Context Diagram, Data Flow Diagrams (Level-0, Level-1 and Level-2), Class Diagram, Sequence Diagram, Activity Diagram, State Diagram, State Machine Diagram, architecture diagram and test cases. The goal is to:

- Clearly define the system boundaries, external entities, and data exchanges.
- Decompose major system functions into detailed processes to understand internal workflows.
- Illustrate object relationships and dynamic behaviours through UML models.
- Provide a logical blueprint of the Netflix system that will guide later design and implementation stages.

Functional & Non-Functional Requirements of Netflix:

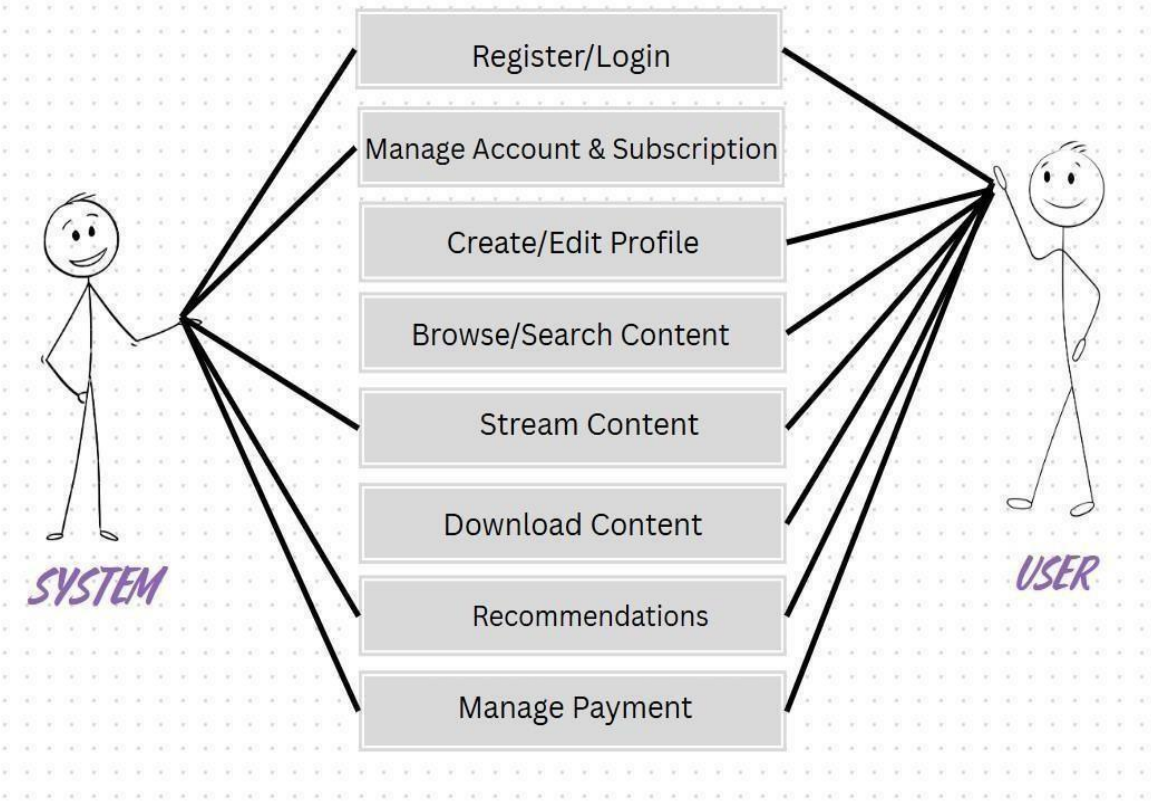
Functional Requirements

- User registration, login & profiles
- Browse & search movies/series
- Play, pause, resume streaming
- Auto-quality adjustment
- Subscription plans & online payments
- Recommendations based on watch history
- Multi-device support (mobile, web, TV)

Non-Functional Requirements

- Fast performance & low buffering
- High availability (99.9% uptime)
- Scalability for millions of users
- Strong security & encrypted data
- Simple, user-friendly interface
- Multi-language support
- Cross-platform compatibility

Use Case:



Use Case Description:

Use Case 1: User Registration/login

Use Case Name	**User Registration/Login**
Actors	User, System
Description	User signs up with email, password, and payment details. The system verifies the information and creates an account. Existing users can sign in to access their accounts.
Trigger	User selects the "Sign Up" or "Sign In" option on the application.

Normal Flow	**For Registration:** 1. User selects "Sign Up". 2. System displays the registration form. 3. User provides a valid email, password, and payment details. 4. System validates the inputs and processes the payment. 5. System creates a new account and stores the information. 6. System automatically logs the user in. **For Login:** 1. User selects "Sign In". 2. System displays the login form. 3. User enters their registered email and password. 4. System authenticates the credentials and grants access.
Alternative Flow	A message is displayed upon the following reasons: 1. **Registration:** Email already exists, passwords are too weak, or payment is declined. 2. **Login:** Incorrect email/password combination.

Use Case 2: Manage Account & Subscription

Use Case Name	**Manage Account & Subscription**
Actors	User, System

Description	The user updates account details such as subscription plan, billing information, or password. The user can also cancel their subscription.
Trigger	User navigates to the "Account" section of the application.
Normal Flow	1. User selects an option to update (e.g., "Change Plan," "Update Billing"). 2. System displays the relevant form/options. 3. User enters and confirms the new information. 4. System validates and saves the changes. 5. System displays a success confirmation message.
Alternative Flow	An error message is displayed if: 1. New payment information is invalid or declined. 2. A new password does not meet the security criteria.

Use Case 3: Create/Edit Profile

Use Case Name	**Create/Edit Profile**
Actors	User, System

Description	The user creates or modifies profiles under their main account, each with unique settings and preferences (e.g., name, language, viewing restrictions).
Trigger	User selects the "Manage Profiles" option.
Normal Flow	1. User selects "Add Profile" or chooses an existing profile to edit. 2. System displays the profile creation/editing form. 3. User enters/modifies the profile name and preferences. 4. User saves the changes. 5. System creates a new profile or updates the existing one.
Alternative Flow	An error message is displayed if: 1. The user tries to add a profile when the maximum number of profiles has been reached. 2. The chosen profile name is already in use on the account.

Use Case 4: Browse/Search Content

Use Case Name	**Browse/Search Content**
Actors	User, System

Description	The user looks for content by navigating through categories (genre, popularity) or by using the search function to find specific titles, actors, or directors.
Trigger	User opens the application or selects the search bar.
Normal Flow	For Browsing:
	1. User scrolls through the content carousels on the main screen. 2. User selects a category to view more titles. For Searching: 1. User enters text into the search bar. 2. System dynamically displays titles matching the query. 3. User selects a title from the results.
Alternative Flow	If a search returns no results, the system displays a "No results found" message and may provide alternative suggestions.

Use Case 5: Stream Content

Use Case Name	**Stream Content**
Actors	User, System
Description	The user selects a movie or show to watch. The system delivers video content to the user's device, adjusting quality based on network speed.
Trigger	User clicks the "Play" icon on a selected title.
Normal Flow	1. User selects a title to watch. 2. System verifies the user's subscription is active. 3. System begins playing video stream. 4. System adapts the streaming quality based on the user's internet bandwidth.
	System saves the viewing progress for the user's profile.

Alternative Flow	1. If the internet connection is poor, the system will buffer or lower the video quality. 2. If the user's subscription is inactive, the system will block playback and prompt the user to resubscribe.

Use Case 6: Download Content

Use Case Name	**Download Content**
Actors	User, System
Description	The user saves a movie or shows to their device to allow for offline viewing at a later time.
Trigger	User clicks the "Download" button on an eligible title.

Normal Flow	1. User finds a downloadable title and clicks "Download". 2. System checks for sufficient device storage. 3. System begins the download process. 4. System notifies the user when the download is complete. 5. The content becomes available in the "Downloads" section.
Alternative Flow	An error message is displayed if: 1. The device does not have enough storage space. 2. The user has reached the maximum limit of downloaded titles. 3. The internet connection is lost (download will pause).

Use Case 7: Recommendation System

Use Case Name	**Recommendation System**
Actors	System (Primary), User (Secondary)
Description	The system analyzes a user's viewing history, ratings, and preferences to suggest new content they are likely to enjoy.

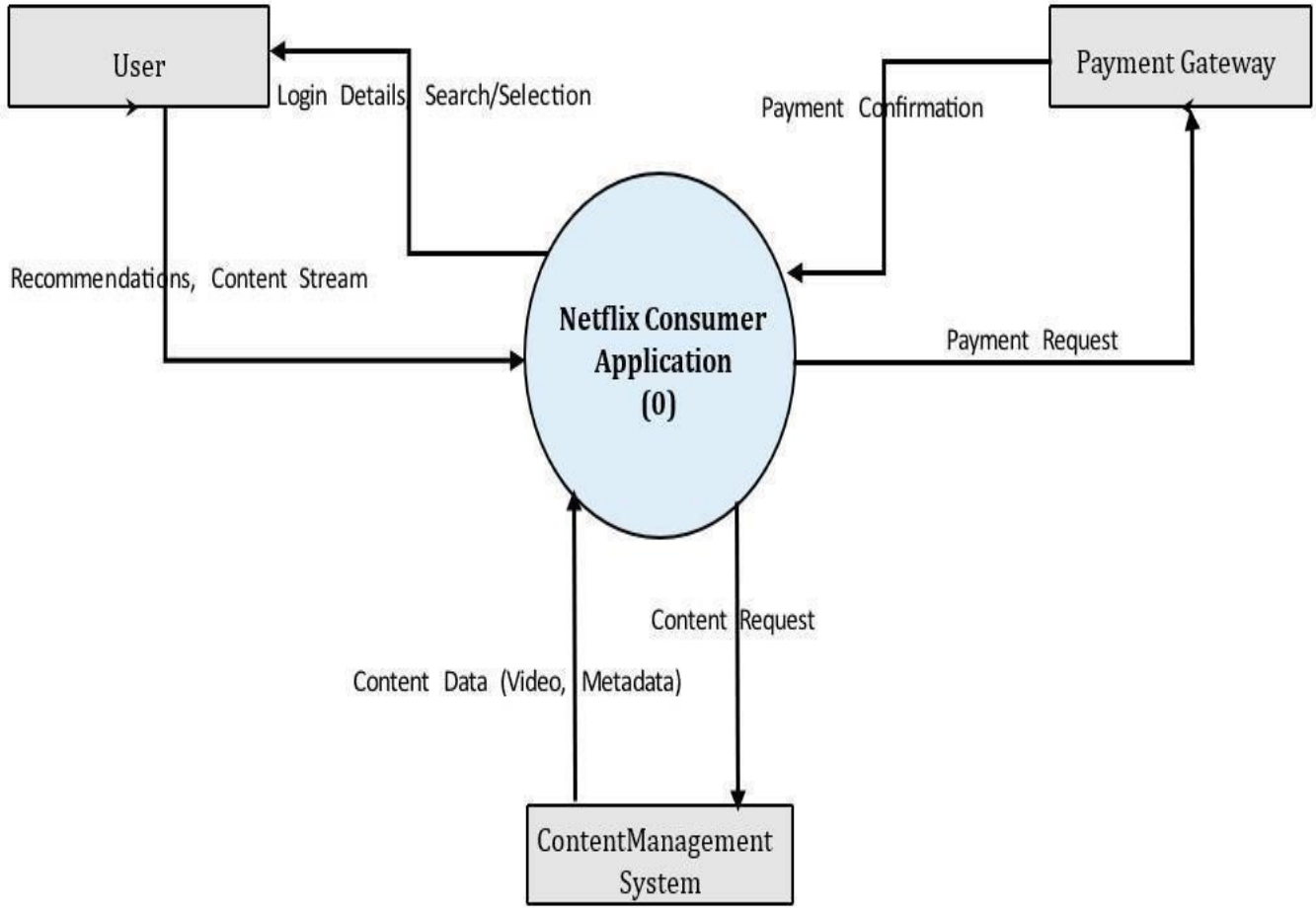
Trigger	This is a continuous background process, with results displayed when the user opens the application's home screen.
Normal Flow	1. User watches and rates content over time. 2. System collects and analyzes this viewing data. 3. System's algorithm generates a list of personalized recommendations. 4. System displays these recommendations in categorized rows on the user's home screen (e.g., "Top Picks for You").
Alternative Flow	For a new user with no viewing history, the system will display globally popular, trendy, or generally acclaimed content instead of personalized recommendations.

Use Case 8: Subscription Payment

Use Case Name	**Subscription Payment**
Actors	User, System
Description	The user provides payment information for a selected subscription plan, which is then processed via a secure payment gateway.

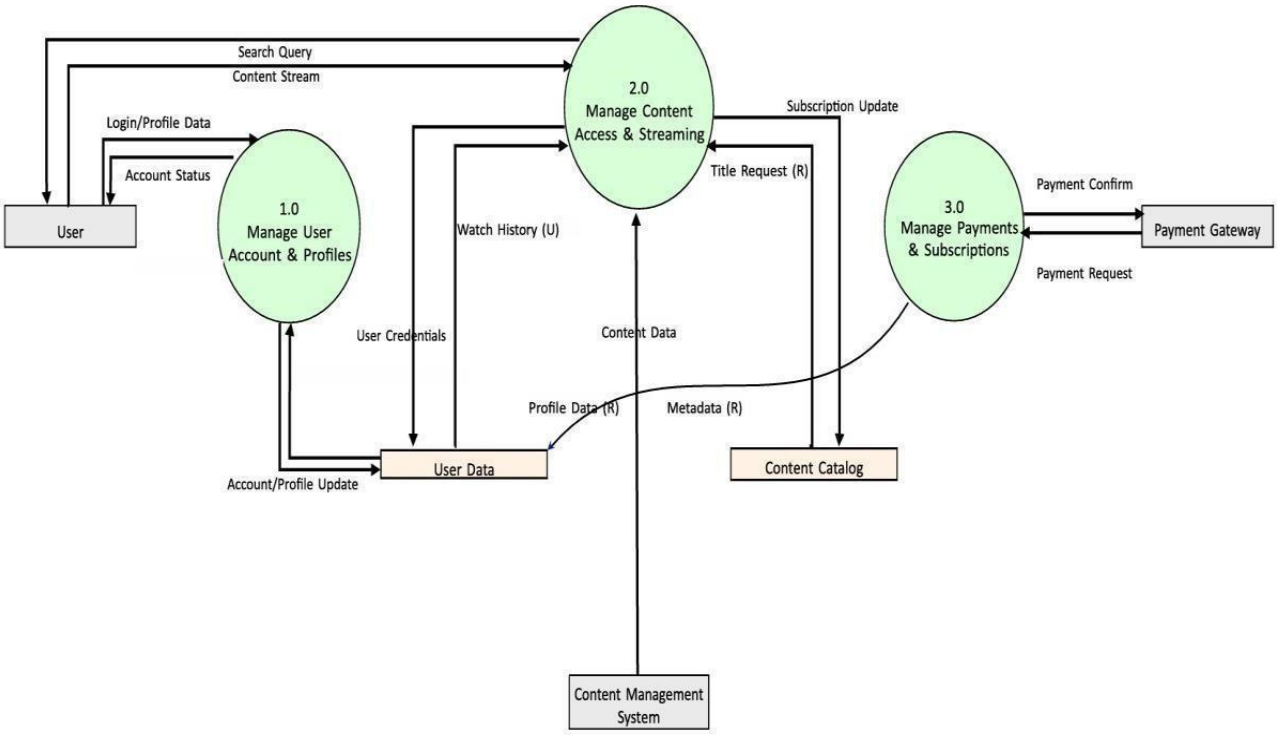
Trigger	User selects a subscription plan during registration or when changing plans.
Normal Flow	**1.** User chooses a subscription plan and proceeds to payment. **2.** System displays the payment form. **3.** Users enter their payment details (e.g., credit card). **4.** System securely transmits the data to the payment gateway. **5.** Payment gateway authorizes the transaction. System activates the subscription and displays confirmation.
Alternative Flow	If the payment gateway declines the transaction (e.g., insufficient funds, incorrect card details), the system displays a payment failure message and asks the user to provide a different payment method.

Context Diagram:



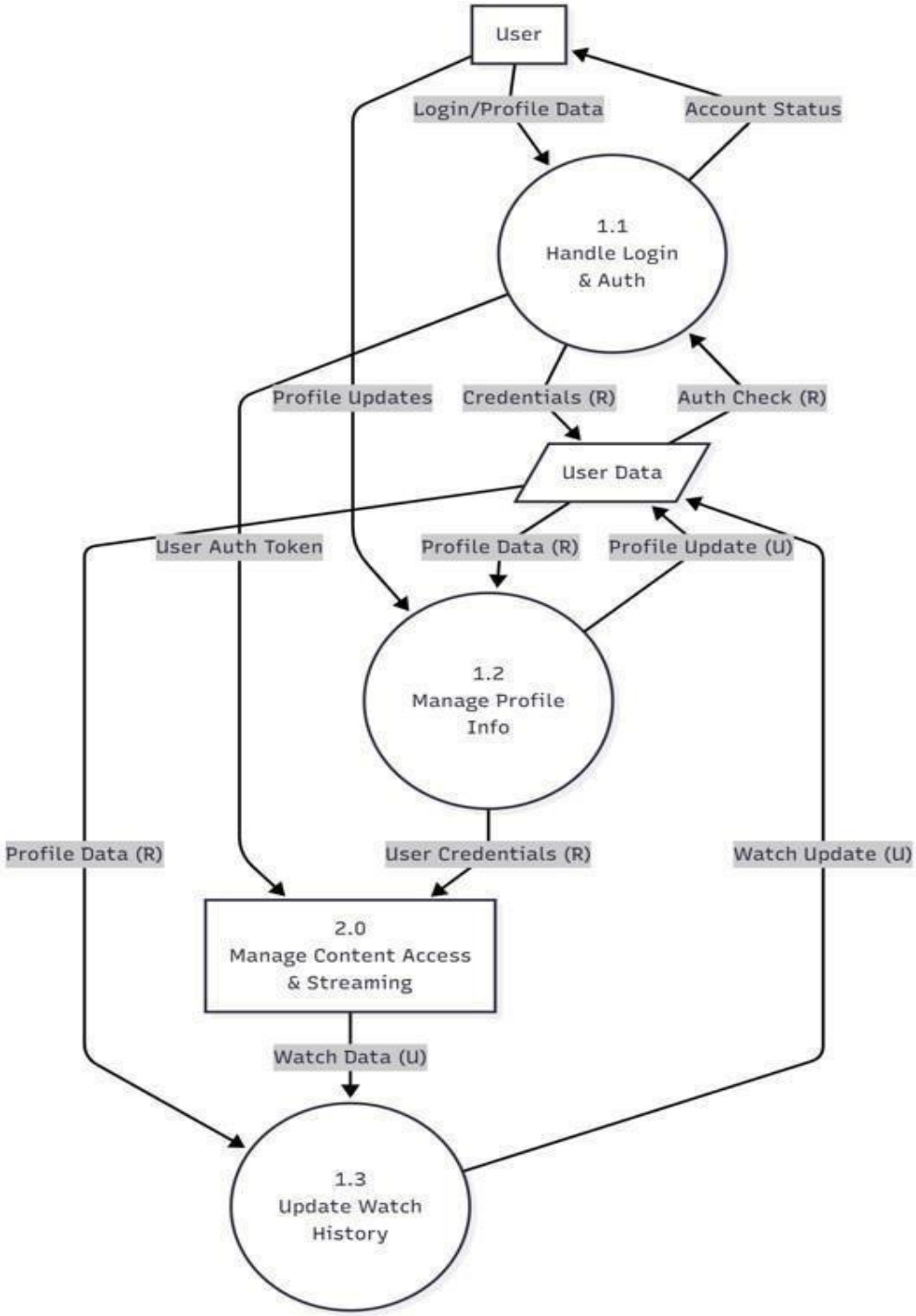
Data Flow Diagram

Level 1:

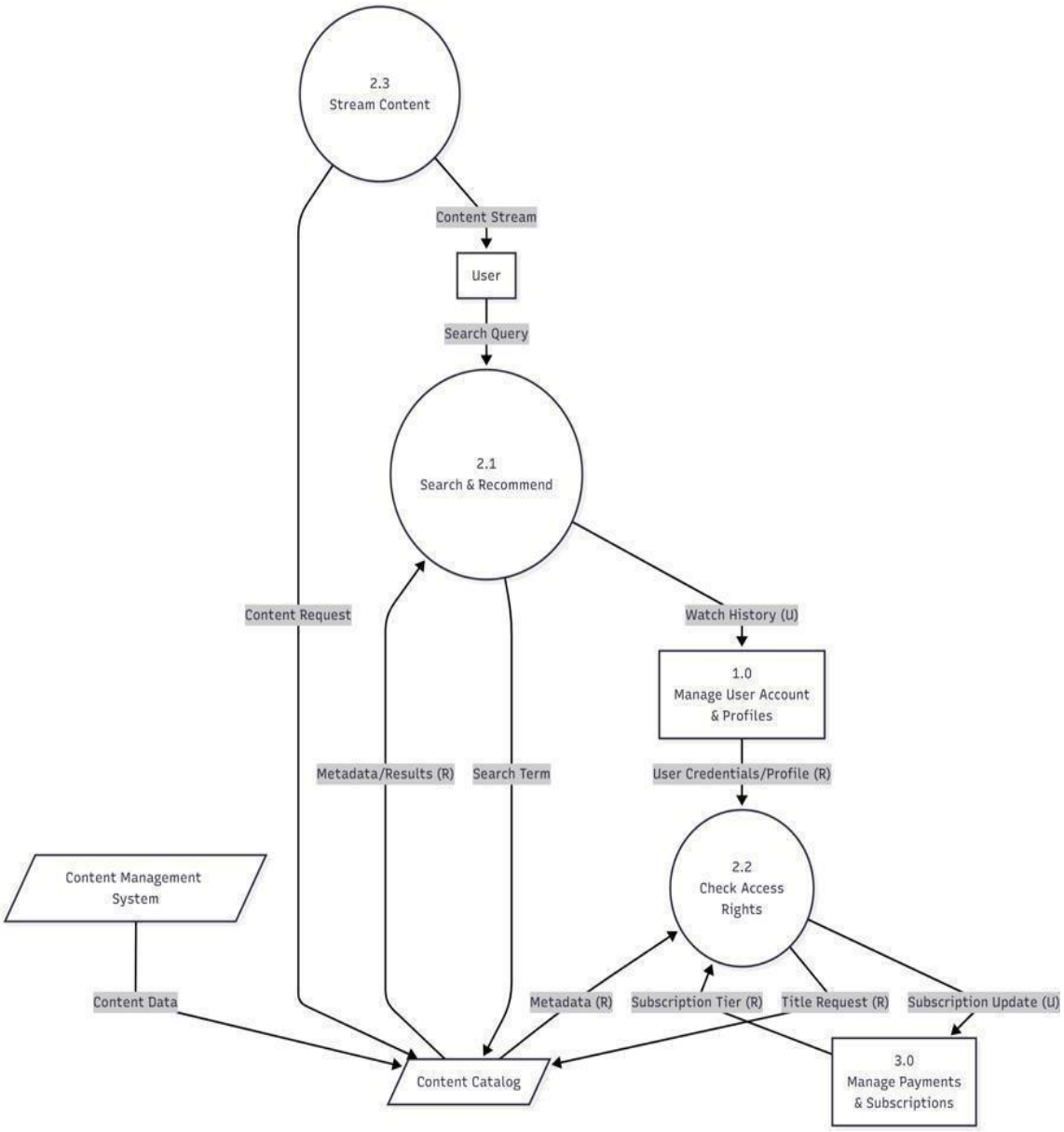


Level 2:

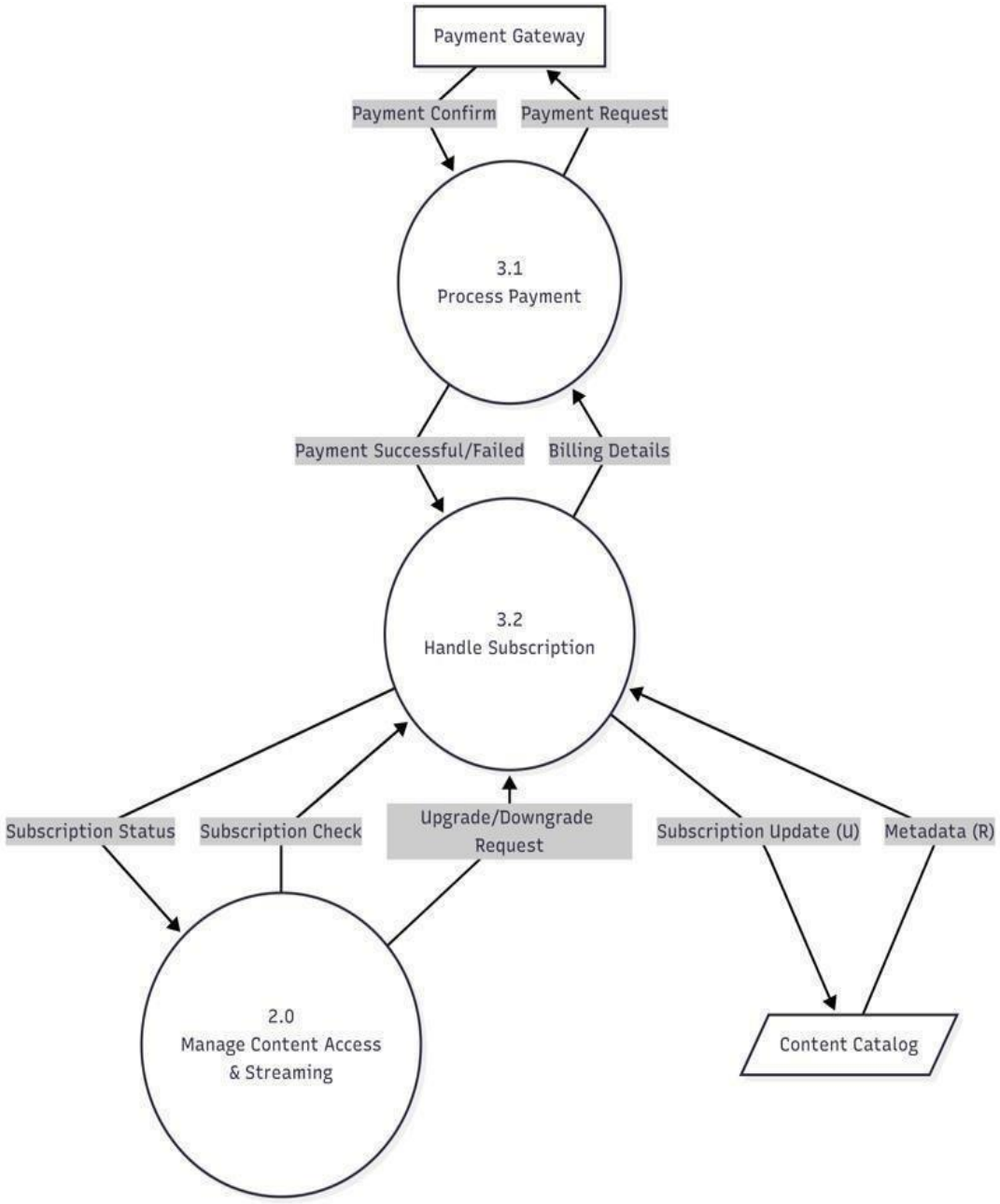
Manage user accounts and profiles:



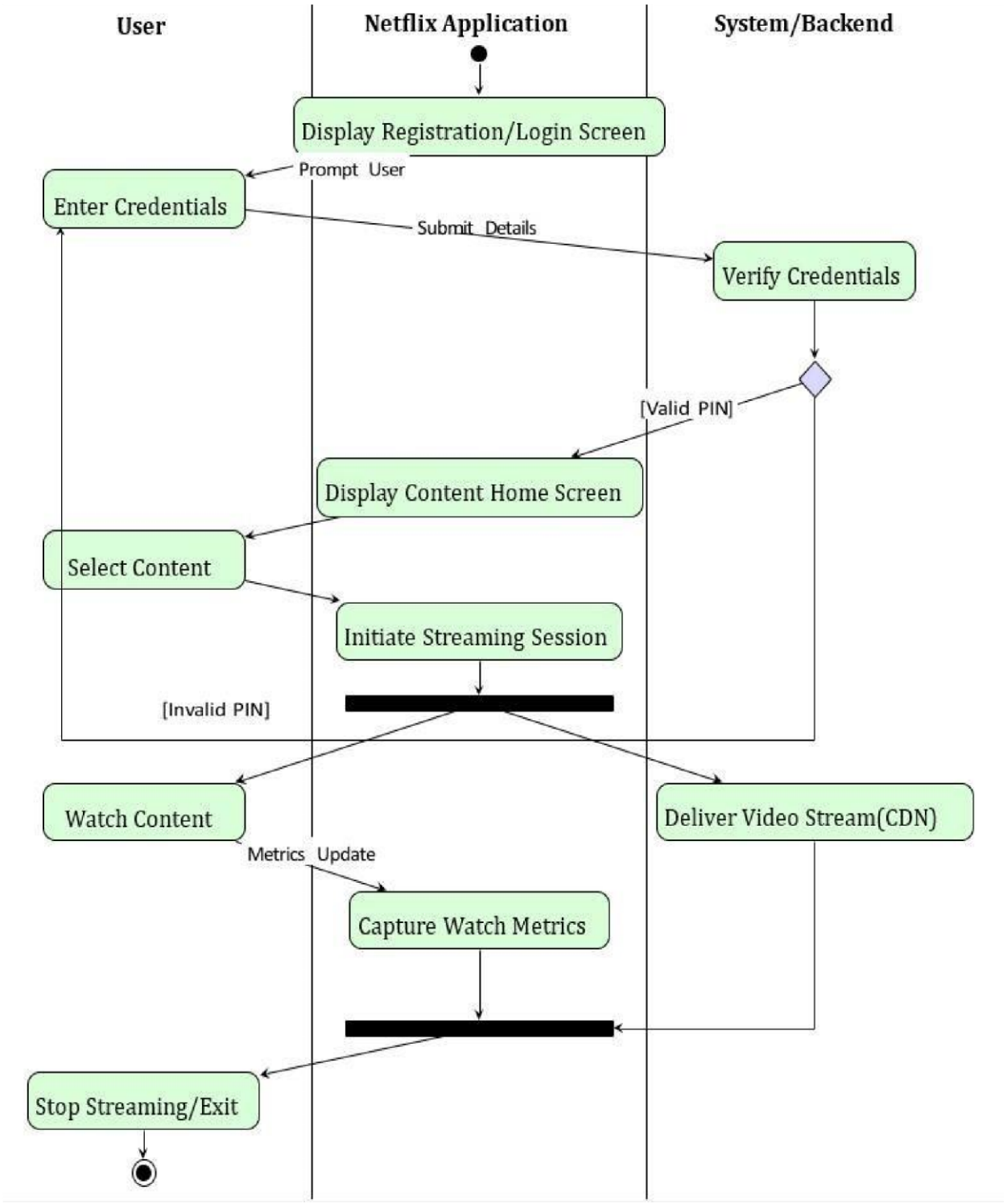
Manage content streaming:



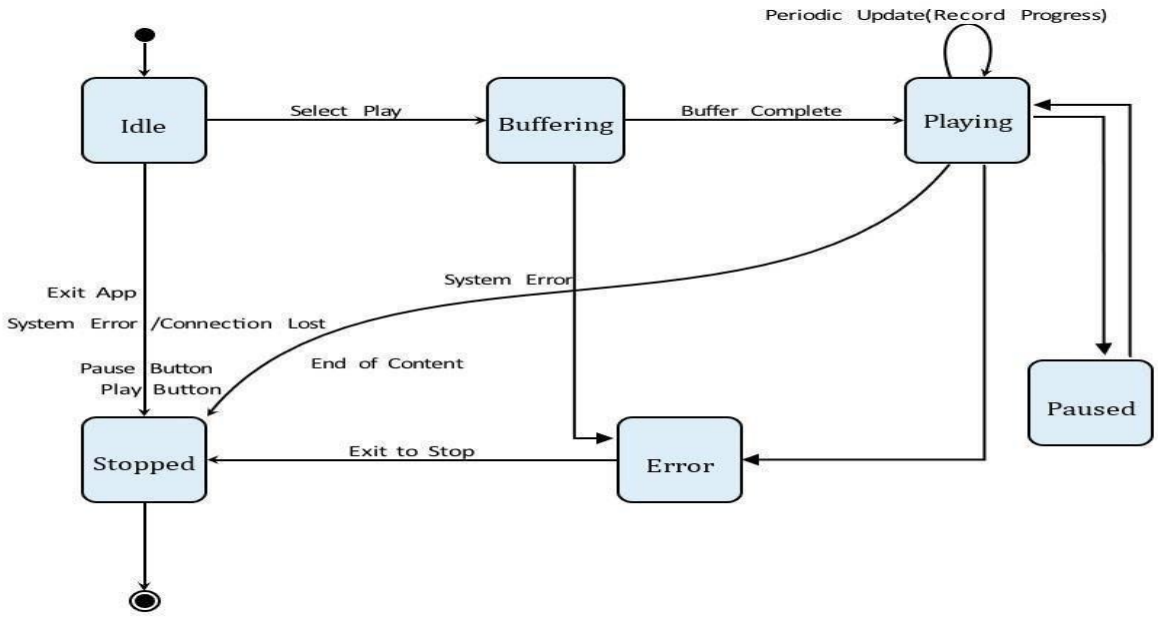
Manage payment and subscriptions:



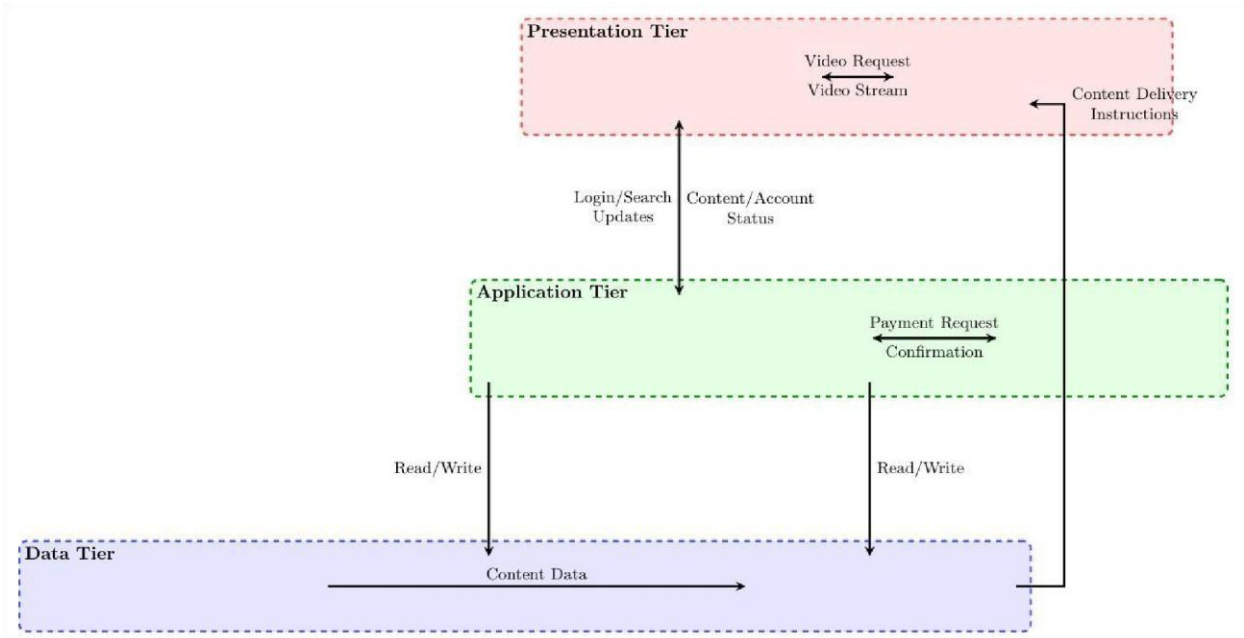
Activity Diagram:



State Flow Diagram:

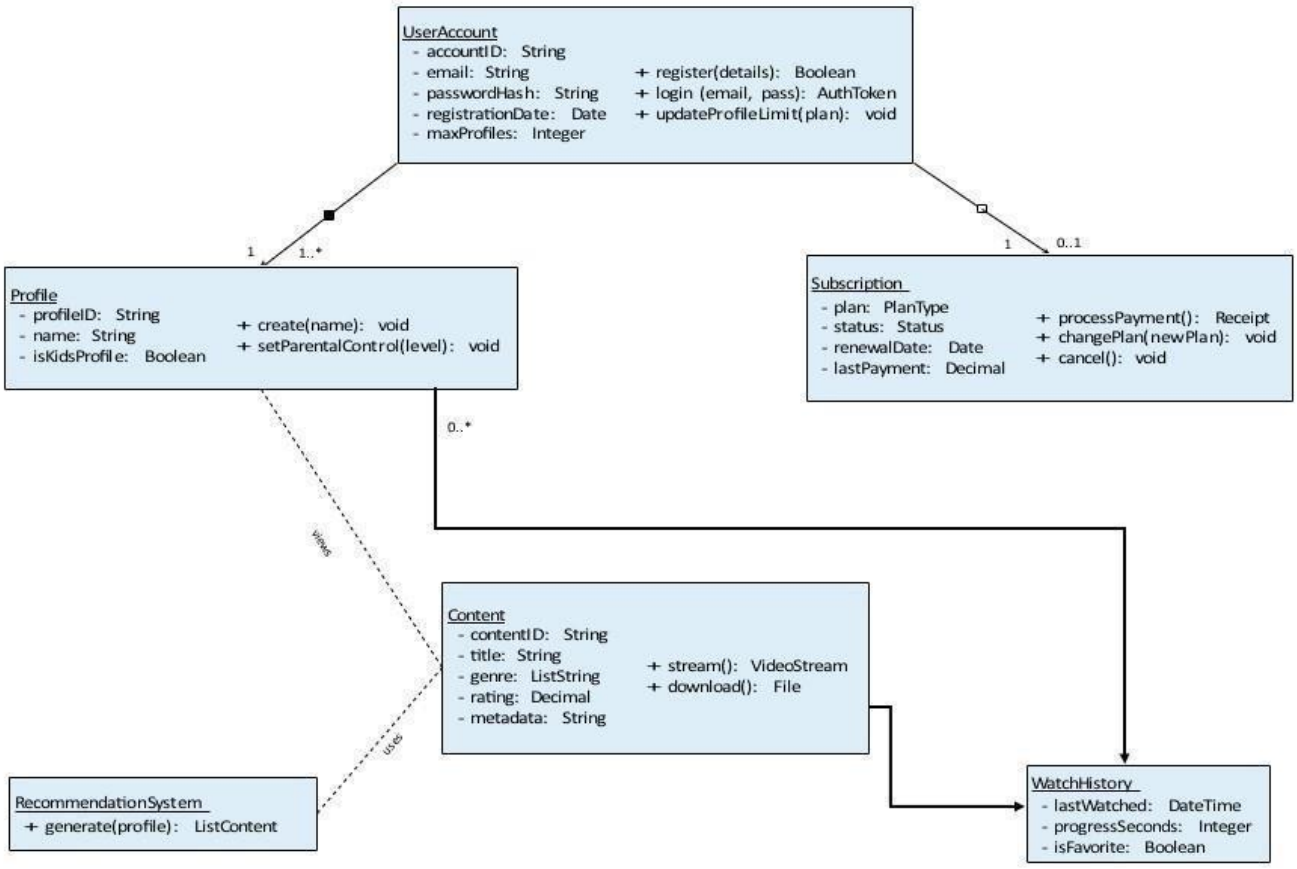


Architecture Diagram:

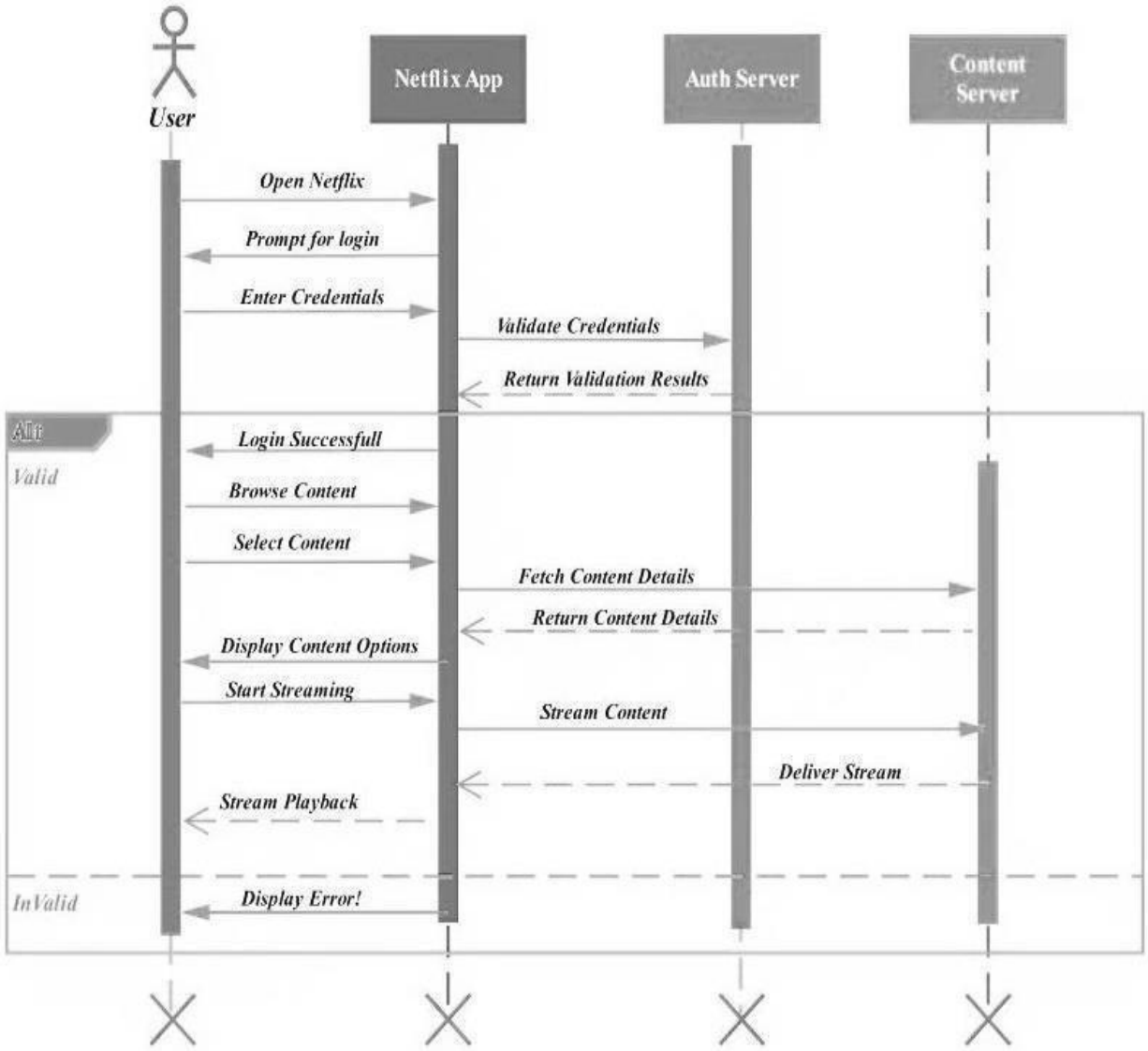


UML

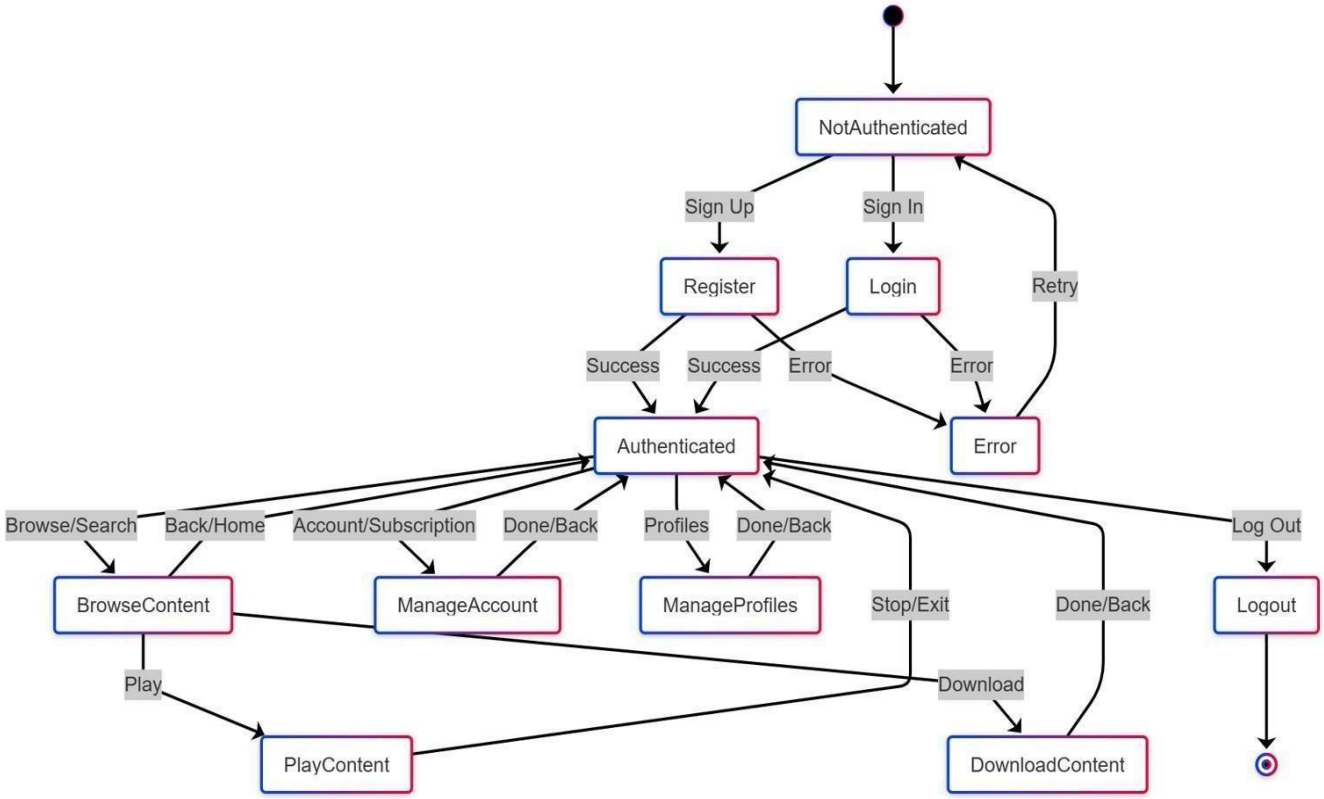
Class Diagram:



Sequence Diagram:



State Machine Diagram:



Test Cases:

Test Case 1: Verify login

Field	Description
Test Case ID	TC_LOGIN_001
Test Case Description	Verify login with invalid password

Pre-Conditions	User must have an existing Netflix account
Test Steps	1. Open Netflix login page 2. Enter valid email 3. Enter invalid password 4. Click “Sign In”
Test Data / Input	Email: user@gmail.com Password: wrongPass123
Expected Result	Error message: **“Incorrect password”** appears and the user should not be logged in
Actual Result	TBD
Status	TBD

Test case 2: Verify Search Functionality

Field	**Description**
Test Case ID	TC_SEARCH_002
Test Case Description	Verify search functionality with valid movie name

Pre-Conditions	User must be logged in
Test Steps	1. Open Netflix homepage 2. Click search bar 3. Enter movie name 4. Press Enter
Test Data / Input	Search: **“Stranger Things”**
Expected Result	Search results related to **“Stranger Things”** should be displayed
Actual Result	TBD
Status	TBD

Test Case 3: Checking MY LIST Function

Field	**Description**
Test Case ID	TC_LIST_003
Test Case Description	Verify adding an item to “My List”
Pre-Conditions	User must be logged in

Test Steps	1. Search for a title 2. Click on the movie/series 3. Click **Add to My List**
Test Data / Input	Title: **Squid Game**
Expected Result	Title should appear in **My List** section
Actual Result	TBD
Status	TBD

White Box Testing (Focus on internal logic/code)

Test Case ID	Feature	White Box Testing Focus
TC_LOGIN_001	Login with valid credentials	Verify internal authentication logic: correct password hashing, email lookup, condition checks for active account, and prevention of SQL injection.
TC_SEARCH_002	Verify search functionality	Validate internal search logic: query parsing and sanitization, matching algorithm against titles/metadata, relevance ranking logic, handling of empty or partial queries, and correct retrieval of results from the database/index.

TC_LIST_003	Add to “My List”	Validate item-saving logic: internal function updates users saved-list array/object, duplication check condition, correct database update call, and retrieval of updated list.

References

McGraw-Hill. (2020). Pressman, R. S. (2020). Software Engineering: A Practitioner’s Approach. McGraw-Hill. *Software Engineering: A Practitioner’s Approach*.

Sommerville, I. (2016). Software Engineering. Pearson Education.